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Computer Networks

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Lab 06

Basic Router Access & CLI Configuration

Department Of Software Engineering

Physical Topology Setup

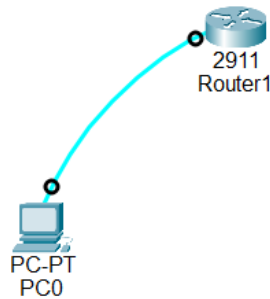


Figure 1 PC0 & Router Connection

Explanation

In this initial step, the physical environment was simulated in Cisco Packet Tracer. A Cisco 2911 router and a standard PC were added to the workspace and connected using a light-blue console (rollover) cable from the PC's RS-232 port to the router's console port.

Navigating IOS Modes

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
```

Explanation

The terminal application on the PC was accessed using default port configurations. The enable command was used to shift from User EXEC mode to Privileged EXEC mode (Router#), and then configure terminal was executed to access Global Configuration mode (Router(config)#) for administrative tasks.

Configuring Device Passwords

```
Router(config)#enable password my_clear_password
Router(config)#service password-encryption
Router(config)#enable secret my_secure_password
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Explanation

Security parameters were applied in global configuration mode. A standard enable password was configured, followed by the activation of the password encryption service. Finally, a highly secure MD5/SHA algorithm-based 'enable secret' password was implemented for strong privilege mode access control.

Verifying the Running Configuration File

```
show running-config
Building configuration...

Current configuration : 794 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$U96zylNGVPG.2soL9BHc40
enable password 7 082C55710A15001600341C05393833272131
!
```

Explanation

The show running-config command was executed in privileged mode to read the device configuration temporarily active in RAM. The output verifies that the clear-text password has successfully changed into ciphertext by the encryption service, while the secret password is natively secured.

Saving Configuration to NVRAM

```
Router# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

```
Router con0 is now available
```

```
Press RETURN to get started.
```

Explanation

To prevent data loss after power cycles, the dynamic running configuration file in RAM was successfully copied into the non-volatile random-access memory (NVRAM) as startup-config using the copy running-config startup-config command.